Lecture 1: LLM and transformer foundations

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Agenda

Predicting the next word

Tokenization turns text into supervised data

Vocabulary
$\mathcal{V} = \{$<pad>, see, spot, run, fast, <eot>$\}$, with IDs $0, \ldots, 5$.

The sentence see spot run fast <eot> becomes $[1, 2, 3, 4, 5]$. **Context-window length: $L = 4$ tokens.** Each prediction uses at most the four most recent token IDs; left-pad shorter prefixes with 0:

prefix	$\mathbf{x}^{(i)} \in \mathcal{X} \subset \{0, \ldots, 5\}^4$	$y^{(i)} \in \mathcal{Y} = \mathcal{V}$
see	$[0, 0, 0, 1]$	spot (2)
see spot	$[0, 0, 1, 2]$	run (3)
see spot run	$[0, 1, 2, 3]$	fast (4)
see spot run fast	$[1, 2, 3, 4]$	<eot> (5)

Supervised learning

The observed dataset is $\mathbf{X}, \mathbf{Y} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=0}^{N-1}$. Training produces a model g whose predictions $g(\mathbf{X}) = \hat{\mathbf{Y}}$ approximate $\mathbf{Y}$.

A one-hidden-layer MLP scores every possible next token

Each ID indexes a learned row of an embedding table $\mathbf{E} \in \mathbb{R}^{6\times 2}$.

For $\mathbf{x} = [x_1, \ldots, x_4]$:

$$\mathbf{E}_{\mathbf{x}} = [\mathbf{E}_{x_1}; \mathbf{E}_{x_2}; \mathbf{E}_{x_3}; \mathbf{E}_{x_4}] \in \mathbb{R}^{4\times 2}, \quad \mathbf{z} = \mathrm{vec}(\mathbf{E}_{\mathbf{x}}) \in \mathbb{R}^8.$$

$$\mathbf{h} = \tanh(\mathbf{W}_1\mathbf{z} + \mathbf{b}_1), \quad \mathbf{s} = \mathbf{W}_2\mathbf{h} + \mathbf{b}_2, \quad \hat{\mathbf{p}} = \mathrm{softmax}(\mathbf{s}).$$

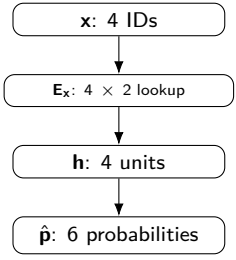
Here $\hat{p}_j = P_g(Y = j \mid \mathbf{x})$; categorical cross-entropy (multiclass log loss) for one example is $-\log \hat{p}_{y^{(i)}}$.

Gradient descent updates $\mathbf{E}, \mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2$.

Loss example: if $y^{(i)} = 4$ and $\hat{p}_4 = 0.80$, then $-\log(0.80) \approx 0.223$.

Layer flow: 4 token IDs $\to$ $4 \times 2 = 8$ embedding coordinates $\to$ 4 `tanh` units $\to$ 6 next-token probabilities.

Each layer's output matches the next layer's input: $(4, 2) \to 8 \to 4 \to 6$.

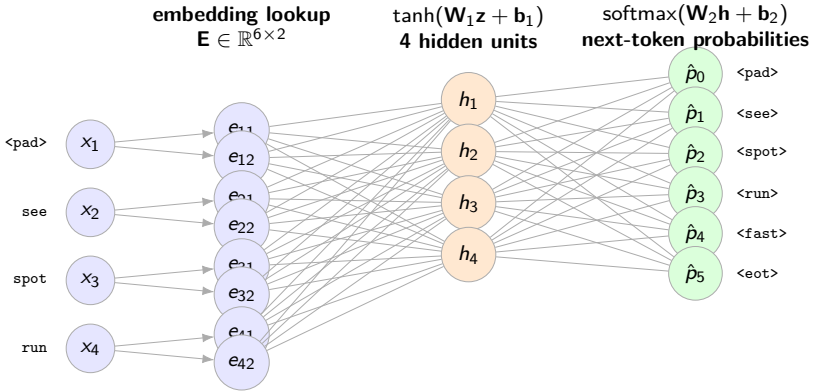


Why not predict ID 4 directly?

IDs are labels, not ordered quantities. Six softmax probabilities define a categorical prediction.

The example MLP

For $\mathbf{x} = [0, 1, 2, 3]$ (<pad>, see, spot, run), the context-window length is $L = 4$: each token contributes two learned coordinates. Every embedding coordinate connects to every hidden unit; every hidden unit scores every vocabulary item.



The output row with the largest probability is decoded; in this training example, $\arg\max_j \hat{p}_j = 4$, or fast.

Detokenization makes the prediction readable

For the context `see spot run`, tokenization gives $\mathbf{x} = [0, 1, 2, 3]$. After the MLP:

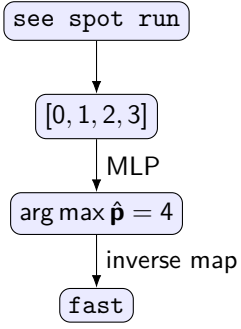
$$\hat{\mathbf{p}} = \left[\hat{p}_{\texttt{<pad>}}, \hat{p}_{\texttt{see}}, \hat{p}_{\texttt{spot}}, \hat{p}_{\texttt{run}}, \hat{p}_{\texttt{fast}}, \hat{p}_{\texttt{<eot>}}\right].$$

Decode

$\hat{y} = \arg\max_{j \in \{0,\ldots,5\}} \hat{p}_j$ is an ID. The inverse vocabulary map converts it back to text: $\hat{y} = 4 \mapsto$ `fast`.

Generate another token

Append the decoded ID, shift the context to $[1, 2, 3, 4]$, and predict again. A learned model should then assign high probability to <eot>.



What the two-dimensional embedding plot shows

The notebook plots the learned 2-D vector $\mathbf{E}_j \in \mathbb{R}^2$ associated with each vocabulary token after training.

- Tokens used in similar predictive contexts can become nearby; this is an empirical pattern, not a guarantee in a tiny dataset.
- The axes themselves have no fixed linguistic interpretation: rotating or reflecting all embeddings can leave equivalent downstream behavior after compensating weights.
- The plot is useful for inspecting the representation, while softmax probabilities remain the model's next-word output.

Notebook correspondence

Same vocabulary, four padded examples, 2-D token embeddings, 4 hidden units, cross-entropy training, decoding, and an embedding scatter plot.

$$\text{text} \rightarrow \text{IDs} \rightarrow \text{embeddings} \rightarrow \text{probabilities} \rightarrow \text{text}$$